

WASP-35b

R-1749

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-35_140112 · created 2026-08-02T19:40:14+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456669.6448 +/- 0.0002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1161 +/- 0.0093
Transit depth (Rp/Rs)^2	1.35 +/- 0.22 [%]
Orbital Inclination (inc)	89.56 +/- 1.01
Airmass coefficient 1 (a1)	1.0006 +/- 0.0095
Airmass coefficient 2 (a2)	0.0016 +/- 0.0069
Scatter in the residuals of the lightcurve fit is	0.95 %
Best Comparison Star	#1 - [345, 170]
Optimal Aperture	3.47
Optimal Annulus	10.22
Transit Duration (day)	0.1231 +/- 0.0012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1161 $\pm$ 0.0093 vs 0.1241 (deviation 0.83 σ)
Duration fit vs published	0.1231 d vs 0.1317 d (deviation 6.92 σ)
Mid-time O-C	0.6 min
Depth SNR	12.1
Residual scatter	0.95 %

Light curve

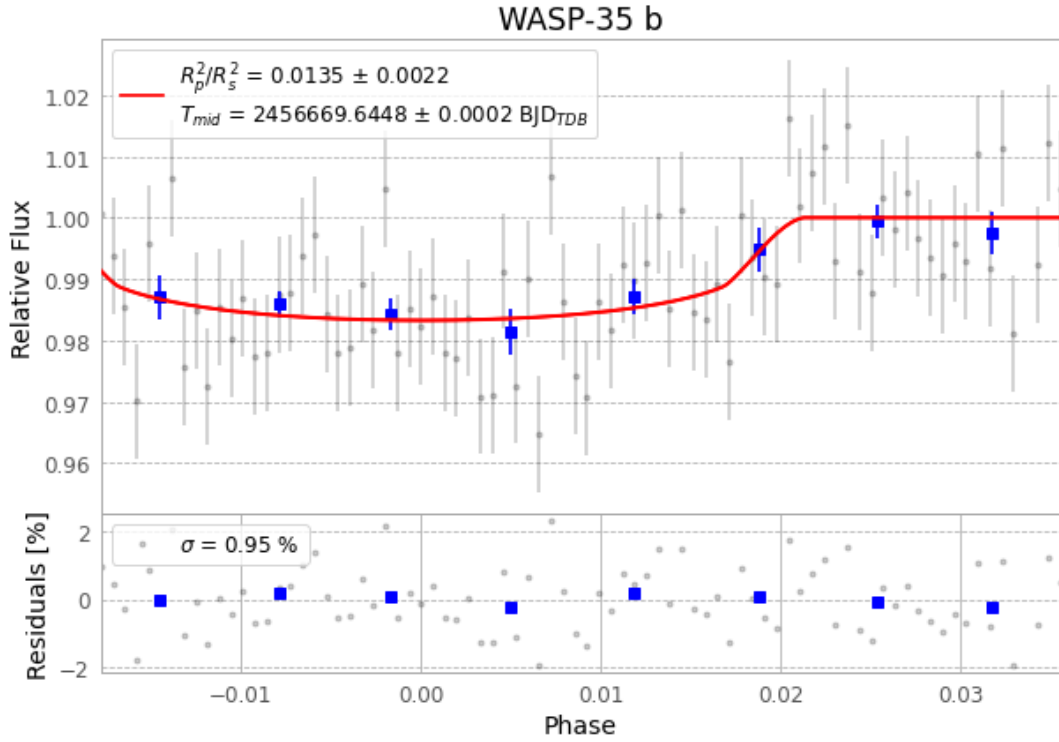


Figure 1 — FinalLightCurve_WASP-35 b_2014-01-11.png (SHA-256 d36d1993e8ec8aa2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [227, 222], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 96dd2d5e5e350979...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

82 FITS frames (54 MB), WASP-35140112020012.FITS → WASP-35140112060312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-35-140112.log (a77347ad2318...), FinalLightCurve_WASP-35 b_2014-01-11.png (d36d1993e8ec...), FinalLightCurve_WASP-35 b_2014-01-11.csv (76d4b004f243...)

Compute resources

Wall time 170.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 19:40Z.